

General Design Notes

Assumed Stackup:
----- Cu 1oz (SIG, PWR / PWR pours)
Fr4
----- Cu 0.5oz (GND plane)
Fr4
----- Cu 0.5oz (GND plane)
Fr4
----- Cu 1oz (SIG, GND pour)

Power Tree:
Dell 0JU012 power supply 19.5V 6.7A
-> stepper motor drivers x 3 to power motors up to 1A per phase (allowance to be 130% rated current to allow for when more than one phase powered) 3.9A
-> some extra 20V headers, no specific power allowance made for them (6.7 - 4.7 = 2A)
-> buck converter 0.8A in (av. in 0.56A), 3A 5V out
-> MCU (ATTiny1616) 0.08A
-> LED's, buttons etc 0.5A
-> 2 x servo motors MG92B (estimated stall current 850mA each <https://forums.adafruit.com/viewtopic.php?t=104598>) 1.7A
-> fan (Noctua NF-A4X10) 0.05A
-> some extra 5V headers for power to random sensors etc, no specific power allowance made for them (3 - 2.33 = 0.67A)

Connectivity:
2x2 male pin headers are only and exclusively used for 20V, so that with corresponding plug housing it is unlikely any non-20V tolerant downstream device will be plugged in to a 20V header.

Mechanical



Sheets

mcu
File: mcu.kicad_sch

stepper_drivers
File: stepper_drivers.kicad_sch

power
File: Power.kicad_sch

Designed by Dave Riley

Sheet: /
File: RobotArmPCB.kicad_sch

Title: Project Normal Robot Arm

Size: A4	Date: 14/09/2025	Rev: 0.2
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Sheet: /mcu/
File: mcu.kicad_sch

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Rev: 0.2
Id: 2/4



